Stochastic Calculus Notes

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1 Random Walks

1. Let ω_i denote the outcome of the ith coin toss. We define,
$$\mathbb{X}_i(\omega_i) = \begin{cases} +1, & \omega_i = H \\ -1, & \omega_i = T \end{cases}$$
2. Also define,
$$\mathbb{M}_n = \sum_{i=1}^{n} \mathbb{X}_i$$
3. Note that, $-n \leq \mathbb{M}_n \leq n$. Also, $\mathbb{M}_n$ and n must have the same parity.
4. Probability that the walker is at $\mathbb{M}_n = k$ after n steps is
$$\mathbb{P}\{\mathbb{M}_n = k\} = p_{n,k} = \frac{1}{2^n}\binom{n}{\frac{n+k}{2}}$$
5. Probability of return to the original position, $k = 0$ after $n = 2\nu$ steps is
$$\mathbb{P}\{\mathbb{M}_{2\nu} = 0\} = \mu_{2\nu} = \binom{2\nu}{v}\frac{1}{2^{2\nu}}$$
6. Number of paths from $(0, 0)$ to (n, x) are,
$$N_{n,x} = \binom{n}{\frac{n+x}{2}}$$
7. **Reflection Principle:** The number of paths from (a, α) to (b, β) with $\alpha, \beta > 0$ which touch or cross the t-axis at least once is equal to the total number of paths from $(a, -\alpha)$ to (b, β).
8. Number of paths from $(0, 0)$ to (n, x) that stay strictly above or below the t-axis can be calculated by taking the first step to $(1, 1)$ and then using the Reflection Princniple. In our case, the paths come out to be
$$N_{n-1,x-1} - N_{n-1,x+1} = \frac{x}{n}N_{n,x}$$
9. Unrelatedly, but obviously, and clearly, $\sum_{k=-n}^{n} N_{n,k} = 2^n$, which is the total possible paths in n time steps.
10. Probability of walker returning to $k = 0$ for the first time in $n = 2\nu$ steps, $f_{2\nu}$ can be calculated from the following relation.
$$\mu_{2\nu} = f_{2\nu}\mu_0 + f_{2\nu-2}\mu_2 + \cdots + f_2\mu_{2\nu-2}$$
11. We can iteratively solve this but let's go for a much more elegant method. For the sake of precision,
$$f_{2\nu} = \mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \ldots, \mathbb{M}_{2\nu-1} \neq 0, \mathbb{M}_{2\nu} = 0\}$$
12. We now prove the following result, which would be instumental in getting a closed form solution for $f_{2\nu}$. Consider the result:
$$\mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \ldots, \mathbb{M}_{2\nu} \neq 0\} = \mathbb{P}\{\mathbb{M}_{2\nu} = 0\} = \mu_{2\nu}$$

Proof. Total number of paths that stay above the t-axis but can be anywhere (> 0) at $n = 2\nu$ is
$$\sum_{l=1}^{\nu}(N_{2\nu-1,2l-1} - N_{2\nu-1,2l+1}) = N_{2\nu-1,1}$$
Also, the probability of getting such paths is
$$p_{2\nu-1,1} = (N_{2\nu-1,1})\frac{1}{2^{2\nu}} = \binom{2\nu-1}{\nu}\frac{1}{2^{2\nu}}$$
To get the expression for our result, we must double the above quantity to account for the paths that stay strictly below the t-axis. Thus,
$$2p_{2\nu-1,1} = \frac{(2\nu)!}{\nu!\nu!}\frac{1}{2^{\nu}} = \mu_{2\nu}$$
□

13. Again consider, $f_{2\nu} = \mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \ldots, \mathbb{M}_{2\nu-1} \neq 0, \mathbb{M}_{2\nu} = 0\}$. All length 2ν paths which satisfy $\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \ldots, \mathbb{M}_{2\nu-2} \neq 0\}$ are $4(2^{2\nu-2}\mu_{2\nu-2}) = 2^{2\nu}\mu_{2\nu-2}$, but this is equal to sum of paths that return to $k = 0$ at 2ν and those that do not. Thus,
$$2^{2\nu}\mu_{2\nu-2} = 2^{2\nu}f_{2\nu} + 2^{2\nu}\mu_{2\nu}$$

14. From the above, we get,
$$f_{2\nu} = \mu_{2\nu-2} - \mu_{2\nu}$$
Also, trivially,
$$f_{2\nu} = \frac{1}{2\nu - 1}\mu_{2\nu}$$

15. Probability of the walker returning back to 0 for the last time at $t = 2l$ in an $n = 2\nu$ step walk is,
$$\alpha_{2l,2\nu} = \mu_{2l} \cdot \mu_{2\nu-2l}$$

16. If ω denotes the sequence of outcomes of a coin toss experminet, and $n_H(\omega)$ and $n_T\omega$ the number of Heads and Tails in that sequence, then
$$\mathbb{P}(\omega) = p^{n_H(\omega)}q^{n_T(\omega)}$$
.

17.
$$\mathbb{E}\{\mathbb{M}_n\} = n(p - q)$$

18. **Markov Property:** The value of a random variable *only* depends on the previous step. $\mathbb{M}_n$ is a Markov Process.

Proof.
$$\mathbb{P}\{\mathbb{M}_n = k \mid \mathbb{M}_{n-1} = a_{n-1}, \ldots, \mathbb{M}_1 = a_1\} = \frac{\mathbb{P}\{\mathbb{M}_n = k, \mathbb{M}_{n-1} = a_{n-1}, \ldots, \mathbb{M}_1 = a_1\}}{\mathbb{P}\{\mathbb{M}_{n-1} = a_{n-1}, \ldots, \mathbb{M}_1 = a_1\}}$$
This evaluates to $\frac{\frac{1}{2^n}}{\frac{1}{2^{n-1}}} = \frac{1}{2}$, but the following quantity also evaluates to the same. Thus,
$$\mathbb{P}\{\mathbb{M}_n = k \mid \mathbb{M}_{n-1} = a_{n-1}, \ldots, \mathbb{M}_1 = a_1\} = \mathbb{P}\{\mathbb{M}_n = k \mid \mathbb{M}_{n-1} = a_{n-1}\}$$
□

19. **Conditional Expectation**
$$\mathbb{E}[\mathbb{X}](\omega_1, \omega_2, \ldots, \omega_n) = \sum_{\substack{\omega \in \Omega_N \\ \text{s.t. } \omega_1,\omega_2,\ldots,\omega_n \text{ fixed} \\ \text{sum over } \omega_{n+1},\omega_{n+2},\ldots,\omega_N}} \mathbb{X}(\omega)\,\mathbb{P}\{\omega \mid \omega_1, \omega_2, \ldots, \omega_n\}$$

20. **Martingale Property:**
$$\mathbb{E}[\mathbb{M}_n \mid \mathcal{F}_m] = \mathbb{M}_m, \quad m \leq n$$

21. **Sub-Martingale:** $\mathbb{E}[\mathbb{M}_n \mid \mathcal{F}_m] \geq \mathbb{M}_m$

22. **Super-Martingale:** $\mathbb{E}[\mathbb{M}_n \mid \mathcal{F}_m] \leq \mathbb{M}_m$

23. $\mathbb{M}_n$ is a martingale. As,
$$\mathbb{E}[\mathbb{M}_n](\omega_1, \omega_2, \ldots, \omega_m) = \mathbb{M}_m(\omega_1, \omega_2, \ldots, \omega_m), \quad for \quad m < n$$

2 Stock Price Process

1. **Binomial Tree Model:** An N-step model where the stock price can either increase or decrease by a constant up or down factor:
$$\mathbb{S}_n = u^{n_H(\omega_1\omega_2\ldots\omega_n)} d^{n_T(\omega_1,\omega_2\ldots\omega_n)} \mathbb{S}_0.$$
For a symmetric model, we have $u = 1/d$, and thus
$$\mathbb{S}_n = u^{\mathbb{M}_n(\omega_1\omega_2\ldots\omega_n)} \mathbb{S}_0.$$

2. **Properties of this Stock Price Process**
   (a) $\mathbb{S}_n$ is an adapted stochastic process.
   (b) $\mathbb{S}_n$ is a Markov process.
   (c) $\mathbb{S}_n$ is a martingale under specific probabilities called the *risk-neutral probabilities*.
   (d) The conditions $\tilde{p}u + \tilde{q}d = 1 + r$ and $\tilde{p} + \tilde{q} = 1$ yield
   $$\tilde{p} = \frac{1+r-d}{u-d}, \quad \tilde{q} = \frac{u-(1+r)}{u-d}.$$
   (e) The discounted stock price process, defined below, is a martingale under the risk-neutral measure:
   $$\mathbb{Y}_n(\omega_1\omega_2\ldots\omega_n) = \frac{\mathbb{S}_n(\omega_1\omega_2\ldots\omega_n)}{(1+r)^n}.$$

3. **Hedging Portfolio:** Suppose a seller of a European call option hedges their risk by purchasing the underlying stock and investing the remainder in the money market. To be fully hedged, the value of the portfolio must equal the payoff of the option in all possible states; otherwise, there is a possibility of arbitrage.

4. The payoff of the call option is $\mathbb{V}_N = \max(\mathbb{S}_N - K, 0)$, and the portfolio evolves as
$$\mathbb{X}_n(\omega_1\ldots\omega_n) = \Delta_{n-1}(\omega_1\ldots\omega_{n-1})\mathbb{S}_n(\omega_1\ldots\omega_n) + (1+r)\big(\mathbb{X}_{n-1}(\omega_1\ldots\omega_{n-1}) - \Delta_{n-1}(\omega_1\ldots\omega_{n-1})\mathbb{S}_{n-1}(\omega_1\ldots\omega_{n-1})\big).$$

5. **For $N = 1$:** $\mathbb{X}_1(\omega_1) = (\mathbb{S}_1(\omega_1) - K)^+$, and
$$\Delta_0 = \frac{\mathbb{X}_1(H) - \mathbb{X}_1(T)}{\mathbb{S}_1(H) - \mathbb{S}_1(T)},$$
$$\mathbb{X}_0 = \frac{1}{1+r}\big[\tilde{p}\,\mathbb{X}_1(H) + \tilde{q}\,\mathbb{X}_1(T)\big] = \tilde{\mathbb{E}}\left(\frac{\mathbb{X}_1}{1+r}\right).$$

6. In general,
$$\Delta_n(\omega_1\ldots\omega_n) = \frac{\mathbb{X}_{n+1}(\omega_1\ldots\omega_nH) - \mathbb{X}_{n+1}(\omega_1\ldots\omega_nT)}{\mathbb{S}_{n+1}(\omega_1\ldots\omega_nH) - \mathbb{S}_{n+1}(\omega_1\ldots\omega_nT)},$$
$$\mathbb{X}_n(\omega_1\ldots\omega_n) = \frac{1}{1+r}\big[\tilde{p}\,\mathbb{X}_{n+1}(\omega_1\ldots\omega_nH) + \tilde{q}\,\mathbb{X}_{n+1}(\omega_1\ldots\omega_nT)\big].$$

7. **Wiener Process**
   (a) It is easy to verify that $\mathbb{E}[\mathbb{M}_n] = 0$ and $\mathbb{E}[\mathbb{M}_n^2] = n$.
   (b) **Scaled Random Walk:** We define
   $$\mathbb{W}^{(p)}(t) = \frac{1}{\sqrt{p}}\,\mathbb{M}_{pt},$$
   where
   $$\mathbb{M}_{pt} = \begin{cases} \mathbb{M}_{pt} & \text{if } pt \text{ is an integer,} \\ \text{linear interpolation} & \text{otherwise.} \end{cases}$$
   (c) The increment of the random walk is $\mathbb{M}_n - \mathbb{M}_m = \sum_{i=m+1}^{n} \mathbb{X}_i$. Clearly, $\mathbb{E}[\mathbb{M}_n - \mathbb{M}_m] = 0$ and $\mathbb{E}[(\mathbb{M}_n - \mathbb{M}_m)^2] = n - m$.
   (d) The Wiener process is defined as the limit
   $$\mathbb{W}(t) = \lim_{p\to\infty} \mathbb{W}^{(p)}(t) = \lim_{p\to\infty} \frac{1}{\sqrt{p}}\,\mathbb{M}_{pt}.$$

(e) The process exhibits fractal (self-similar) behaviour and Hölder continuity. The scaling factor $\sqrt{p}$ is chosen so that the quadratic variation of the Wiener process equals t.

(f) **Properties of the Wiener Process**

i. $\mathbb{W}(t) \sim N(0,\, t)$.

ii. For $0 < t_1 < t_2 < \cdots < t_n$, the increments $\mathbb{W}(t_i) - \mathbb{W}(t_{i-1})$ are independent.

iii. $\mathbb{W}(t_i) - \mathbb{W}(t_{i-1}) \sim N(0,\, t_i - t_{i-1})$.

(g) Since $\mathbb{W}(t)$ is normally distributed, we have

$$\mathbb{P}\{\mathbb{W}(t) \le \omega\} = \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{\omega} e^{-x^2/(2t)}\, dx.$$

8. **Continuous Limit of the Binomial Asset Pricing Model**

The scaled stock price is given by

$$\mathbb{S}_{pt} = u^{n_H} d^{n_T} \mathbb{S}_0 = u^{(pt+\mathbb{M}_{pt})/2} d^{(pt-\mathbb{M}_{pt})/2} \mathbb{S}_0.$$

Choosing u and d such that $ud = 1$ and, in the limit as $p \to \infty$,

$$u = 1 + \frac{\sigma}{\sqrt{p}}, \qquad d = 1 - \frac{\sigma}{\sqrt{p}},$$

we obtain

$$\mathbb{S}_{pt} = \left(1 + \frac{\sigma}{\sqrt{p}}\right)^{(pt+\mathbb{M}_{pt})/2} \left(1 - \frac{\sigma}{\sqrt{p}}\right)^{(pt-\mathbb{M}_{pt})/2} \mathbb{S}_0.$$

Taking $p \to \infty$ yields

$$\lim_{p\to\infty} \mathbb{S}_{pt} = \mathbb{S}(t) = \mathbb{S}_0 \exp\left(\sigma \mathbb{W}(t) - \frac{1}{2}\sigma^2 t\right),$$

which is the *geometric Brownian motion*.

9. Recall that $\mathbb{E}\big[(\mathbb{W}(t_i) - \mathbb{W}(t_{i-1}))^2\big] = t_i - t_{i-1}$, which informally gives $\mathbb{E}[(\Delta\mathbb{W})^2] = \Delta t$, and hence $d\mathbb{W}^2 = dt$. Consequently, $d\mathbb{W}/dt \sim 1/\sqrt{dt}$, which diverges as $dt \to 0$; the Wiener process is therefore nowhere differentiable in the classical sense. To carry out calculus with $d\mathbb{W}$, we must develop a new framework known as *Itô calculus*.

3 Ito Calculus

1. **Filtration of σ-algebras:** Given a random variable $\mathbb{X}$, $\sigma(\mathbb{X})$ is the collection of all sets of the form $\{\mathbb{X} \in B\}$, with B a Borel subset of $\mathbb{R}$. For instance, $\sigma(\mathbb{M}_2) = \{\{HH\}, \{HT, TH\}, \{TT\}, \dots\}$ (together with their unions and complements), and $\sigma(\mathbb{M}_2) \subseteq \mathcal{F}_2$. We say $\mathbb{X}$ is *independent* of a σ-algebra $\mathcal{F}$ if $\mathbb{P}\{A \cap B\} = \mathbb{P}\{A\}\mathbb{P}\{B\}$ for all $A \in \sigma(\mathbb{X})$, $B \in \mathcal{F}$.

2. A sequence (or continuous family) of σ-algebras $\mathcal{F}(t)$ such that $\mathcal{F}(s) \subseteq \mathcal{F}(t)$ for $s \leq t$ is called a *filtration*. It encodes the information revealed up to time t, which only ever grows.

3. **Caution:** $\sigma(\mathbb{M}_n)$ for $n = 1, 2, \dots$ is *not* a filtration; the correct one is $\mathcal{A}_n = \sigma(\mathbb{M}_1, \mathbb{M}_2, \dots, \mathbb{M}_n)$. Similarly, $\sigma(\mathbb{W}(t))$ is not a filtration, but
$$\mathcal{F}(t) = \sigma(\mathbb{W}(s) : 0 \leq s \leq t)$$
is. We say a process is *adapted* to $\mathcal{F}(t)$ if $\mathbb{X}(t)$ is $\mathcal{F}(t)$-measurable for all t.

4. **The Wiener Process is a Martingale.** Let $\mathcal{F}(s)$ be a filtration to which $\mathbb{W}$ is adapted, and take $s \leq t$. Then

   (A) $\mathbb{W}(t) - \mathbb{W}(s)$ is independent of $\mathcal{F}(s)$,

   (B) $\mathbb{W}(s)$ is $\mathcal{F}(s)$-measurable.

   Proof.
   $$\mathbb{E}[\mathbb{W}(t) \mid \mathcal{F}(s)] = \mathbb{E}[\mathbb{W}(t) - \mathbb{W}(s) \mid \mathcal{F}(s)] + \mathbb{E}[\mathbb{W}(s) \mid \mathcal{F}(s)] = \mathbb{E}[\mathbb{W}(t) - \mathbb{W}(s)] + \mathbb{W}(s) = \mathbb{W}(s)$$
   where the first term loses its conditioning by (A) and the second is unchanged by (B). □

5. **Quadratic Variation:** For a function f continuous on $[0, T]$ and a partition $\Pi : 0 = x_0 < x_1 < \cdots < x_n = T$ with mesh $\|\Pi\| = \max_i(x_i - x_{i-1})$, define
$$[f, f](T) = \lim_{\|\Pi\| \to 0} \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big)^2.$$

6. **Quadratic variation of a smooth function is zero.** If f is continuously differentiable, then $f(x_i) - f(x_{i-1}) \approx f'(x_{i-1})(x_i - x_{i-1})$ and

   Proof.
   $$[f, f](T) = \lim_{\|\Pi\| \to 0} \sum_{i=1}^{n} |f'(x_{i-1})|^2 (x_i - x_{i-1})^2 \leq \lim_{\|\Pi\| \to 0} \|\Pi\| \sum_{i=1}^{n} |f'(x_{i-1})|^2 (x_i - x_{i-1}) = \lim_{\|\Pi\| \to 0} \|\Pi\| \int_0^T |f'(x)|^2 \, dx = 0$$
   since the integral is bounded. □

7. **Quadratic variation of the Wiener process.** Consider the scaled random walk $\mathbb{W}^{(p)}(t) = \frac{1}{\sqrt{p}} \mathbb{M}_{pt}$ and a partition $0 = t_0 < t_1 < \cdots < t_n = T$ with t_i chosen such that $pt_i = i$. Then $\mathbb{W}^{(p)}(t_i) - \mathbb{W}^{(p)}(t_{i-1}) = \frac{1}{\sqrt{p}} \mathbb{X}_i$, and so
$$[\mathbb{W}^{(p)}, \mathbb{W}^{(p)}](T) = \sum_{i=1}^{n} \frac{1}{p} \mathbb{X}_i^2 = \frac{n}{p} = T.$$
   Passing to the limit $p \to \infty$,
$$\boxed{[\mathbb{W}, \mathbb{W}](T) = T}$$
   Unlike smooth functions, the Wiener process has non-zero quadratic variation. Informally, $d\mathbb{W}\, d\mathbb{W} = dt$. (This is the reason for choosing the scaling $1/\sqrt{p}$.)

8. **Why do we need non-zero quadratic variation?** To ensure that stock prices are not predictable; a stock with zero quadratic variation would be an ordinary differentiable function of time, and its future would be known, leading to arbitrage opportunities.

9. **Geometric Brownian Motion:** Recall the scaling limit of the binomial model,
$$\mathbb{S}(t) = \mathbb{S}_0 \exp\big(\sigma \mathbb{W}(t) - \tfrac{1}{2}\sigma^2 t\big).$$
   Since $d\mathbb{W}/dt$ does not exist for any t almost surely, we cannot differentiate this the usual way.

10. **Ito-Doeblin Formula (for $f(\mathbb{W}(t))$).** We want to compute $f(\mathbb{W}(T)) - f(\mathbb{W}(0))$. Write it as a telescoping sum over a partition and Taylor expand each term:

$$f(\mathbb{W}(t_i)) - f(\mathbb{W}(t_{i-1})) = f'(\mathbb{W}(t_{i-1}))\big[\mathbb{W}(t_i) - \mathbb{W}(t_{i-1})\big] + \frac{1}{2!}f''(\mathbb{W}(t_{i-1}))\big[\mathbb{W}(t_i) - \mathbb{W}(t_{i-1})\big]^2 + \dots$$

As $\|\Pi\| \to 0$, the higher order terms vanish, but the second order term does *not*, since $\sum_i [\mathbb{W}(t_i) - \mathbb{W}(t_{i-1})]^2 \simeq \sum_i (t_i - t_{i-1}) = T$. Replacing the squared increments by $(t_i - t_{i-1})$,

$$f(\mathbb{W}(T)) - f(\mathbb{W}(0)) = \int_0^T f'(\mathbb{W}(u))\, d\mathbb{W}(u) + \frac{1}{2}\int_0^T f''(\mathbb{W}(u))\, du.$$

In differential form,

$$\boxed{df(\mathbb{W}(t)) = f'(\mathbb{W}(t))\, d\mathbb{W}(t) + \frac{1}{2}f''(\mathbb{W}(t))\, dt}$$

The extra $\frac{1}{2}f''\,dt$ term is the new ingredient compared to the ordinary chain rule. Mnemonic: $d\mathbb{W}\, d\mathbb{W} = dt$.

11. **Ito-Doeblin Formula for $f(t, \mathbb{W}(t))$:**

$$df(t, \mathbb{W}(t)) = \partial_t f(t, \mathbb{W}(t))\, dt + \partial_x f(t, \mathbb{W}(t))\, d\mathbb{W}(t) + \frac{1}{2}\partial_x^2 f(t, \mathbb{W}(t))\, dt.$$

12. **Example (GBM):** Let $\mathbb{S}(t, x) = \mathbb{S}_0 \exp\big(\sigma x - \frac{1}{2}\sigma^2 t\big)$. Then $\partial_t \mathbb{S} = -\frac{1}{2}\sigma^2 \mathbb{S}$, $\partial_x \mathbb{S} = \sigma \mathbb{S}$ and $\partial_x^2 \mathbb{S} = \sigma^2 \mathbb{S}$. Applying the formula, the dt terms cancel:

$$d\mathbb{S}(t, \mathbb{W}(t)) = -\tfrac{1}{2}\sigma^2 \mathbb{S}\, dt + \sigma \mathbb{S}\, d\mathbb{W}(t) + \tfrac{1}{2}\sigma^2 \mathbb{S}\, dt = \sigma \mathbb{S}(t, \mathbb{W}(t))\, d\mathbb{W}(t).$$

13. **GBM with drift:** Let

$$\mathbb{S}(t, \mathbb{W}(t)) = \mathbb{S}_0 \exp\big(\sigma \mathbb{W}(t) + \big(\alpha - \tfrac{1}{2}\sigma^2\big)\, t\big)\,,$$

where σ is the *volatility factor* and α is the *drift* (systematic, non-random). Here $\partial_t \mathbb{S} = (\alpha - \frac{1}{2}\sigma^2)\mathbb{S}$, $\partial_x \mathbb{S} = \sigma \mathbb{S}$, $\partial_x^2 \mathbb{S} = \sigma^2 \mathbb{S}$, so

$$\boxed{d\mathbb{S}(t) = \alpha\, \mathbb{S}(t)\, dt + \sigma\, \mathbb{S}(t)\, d\mathbb{W}(t)}$$

which is the *stochastic differential equation* for the stock price.

14. **Discounted stock price:** Similarly, for the discounted process,

$$d\big(e^{-rt}\mathbb{S}(t)\big) = (\alpha - r)\, e^{-rt}\mathbb{S}(t)\, dt + \sigma\, e^{-rt}\mathbb{S}(t)\, d\mathbb{W}(t).$$

Here r is the continuously compounded interest rate, $r_\infty = \ln(1 + r_1)$ where r_1 is the one-period rate; we simply call it r.

15. **Wealth Equation:** For a portfolio holding Δ_n shares and the rest in the money market, we had

$$\mathbb{X}_{n+1} - \mathbb{X}_n = \Delta_n(\mathbb{S}_{n+1} - \mathbb{S}_n) + r_1\big(\mathbb{X}_n - \Delta_n \mathbb{S}_n\big).$$

In the continuous limit,

$$d\mathbb{X}(t) = \Delta(t)\, d\mathbb{S}(t) + r\big(\mathbb{X}(t) - \Delta(t)\mathbb{S}(t)\big)dt,$$

and substituting $d\mathbb{S}$,

$$d\mathbb{X}(t) = \sigma\Delta(t)\mathbb{S}(t)\, d\mathbb{W}(t) + (\alpha - r)\Delta(t)\mathbb{S}(t)\, dt + r\mathbb{X}(t)\, dt.$$

Using the product rule, the discounted wealth satisfies $d\big(e^{-rt}\mathbb{X}(t)\big) = \Delta(t)\, d\big(e^{-rt}\mathbb{S}(t)\big)$.

16. **Black-Scholes-Merton Equation.** Let $c(t, x)$ be the no-arbitrage price of a European call at time t when the stock price is x. The assumption is that c depends only on the current time and the current stock price. It is a non-random function; the randomness enters only through plugging in $x = \mathbb{S}(t)$.

    (a) By the no-arbitrage theorem, a hedging portfolio that replicates the option must satisfy $e^{-rt}c(t, \mathbb{S}(t)) = e^{-rt}\mathbb{X}(t)$ for all $t \geq 0$, and hence $d\big(e^{-rt}c(t, \mathbb{S}(t))\big) = d\big(e^{-rt}\mathbb{X}(t)\big)$.

    (b) Applying Ito-Doeblin, with $d\mathbb{S}(t)\, d\mathbb{S}(t) = \sigma^2 \mathbb{S}^2\, dt$ (we keep only terms of order $d\mathbb{W}$ or dt; note we write $d\mathbb{S}\, d\mathbb{S}$ and not dt, since $d\mathbb{S}\, d\mathbb{S} \neq dt$),

$$dc = \partial_t c\, dt + \partial_x c\, d\mathbb{S} + \frac{1}{2}\partial_x^2 c\, \sigma^2 \mathbb{S}^2\, dt.$$

(c) Differentiating the discounted option price,

$$d\big(e^{-rt}c\big) = e^{-rt}\left(-rc + \partial_t c + \alpha\mathbb{S}\partial_x c + \frac{1}{2}\sigma^2\mathbb{S}^2\partial_x^2 c\right)dt + e^{-rt}\sigma\mathbb{S}\partial_x c\, d\mathbb{W}.$$

(d) The right hand side is

$$d\big(e^{-rt}\mathbb{X}\big) = e^{-rt}\Delta(t)(\alpha - r)\mathbb{S}\, dt + e^{-rt}\Delta(t)\sigma\mathbb{S}\, d\mathbb{W}.$$

(e) **Comparing the $d\mathbb{W}$ terms:** $\sigma\mathbb{S}\partial_x c = \sigma\mathbb{S}\Delta(t)$, so

$$\Delta(t) = \partial_x c(t, \mathbb{S}(t)),$$

the continuous version of the delta hedging formula.

(f) **Comparing the dt terms:** using $\Delta = \partial_x c$, the α terms cancel and

$$-rc + \partial_t c + \frac{1}{2}\sigma^2\mathbb{S}^2\partial_x^2 c = -r\mathbb{S}\,\partial_x c.$$

This must hold for every possible value of $\mathbb{S}(t)$, so we get the PDE

$$\boxed{\partial_t c + rx\,\partial_x c + \frac{1}{2}\sigma^2 x^2\partial_x^2 c = rc}$$

This is a PDE for the non-random function $c(t,x)$ with parameters r, σ. The drift α does not appear, because we used the delta hedging formula.

17. **Terminal and Boundary Conditions.**

(a) *Terminal condition* from the payoff at $t = T$: $c(T,x) = \max(x - K, 0) = (x-K)^+$.

(b) *At $x = 0$:* Setting $x = 0$ in the PDE gives $\partial_t c(t,0) = rc(t,0)$, so $c(t,0) = c(0,0)e^{rt}$. But $c(T,0) = 0$, so $c(0,0) = 0$ and

$$c(t,0) = 0.$$

(c) *As $x \to \infty$:* $c(t,x)$ may grow at most as a positive power of x, and in fact

$$c(t,x) \sim x - e^{-r(T-t)}K, \quad x \to \infty,$$

which comes from comparing with the value of a forward contract at delivery price K.

18. **Black-Scholes-Merton Formula:** The PDE has the analytical solution

$$c(t,x) = x\,N\big(d_+(T-t,x)\big) - Ke^{-r(T-t)}N\big(d_-(T-t,x)\big),$$

where

$$N(y) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{y} e^{-z^2/2}\,dz, \qquad d_\pm(\tau,x) = \frac{1}{\sigma\sqrt{\tau}}\left[\ln\frac{x}{K} + \left(r \pm \frac{\sigma^2}{2}\right)\tau\right].$$

19. **Checking the terminal condition:** As $t \to T$ (i.e. $\tau \to 0$), $d_\pm$ blows up.

- If $x > K$: $\ln(x/K) > 0$, so $d_\pm \to +\infty$, $N \to 1$, and $c \to x - K$.
- If $x < K$: $\ln(x/K) < 0$, so $d_\pm \to -\infty$, $N \to 0$, and $c \to 0$.

This recovers $(x-K)^+$.

20. **The Greeks:** The derivatives $\partial_x c$, $\partial_t c$, $\partial_x^2 c$ have names.

$$\Delta = \partial_x c = N\big(d_+(T-t,x)\big)$$
$$\Theta = \partial_t c = -rKe^{-r(T-t)}N\big(d_-(T-t,x)\big) - \frac{\sigma x}{2\sqrt{T-t}}N'\big(d_+(T-t,x)\big)$$
$$\Gamma = \partial_x^2 c = \frac{1}{\sigma x\sqrt{T-t}}N'\big(d_+(T-t,x)\big)$$

Clearly $\Delta > 0$ (long position in the stock to hedge a short call), $\Theta < 0$ (at a fixed stock price, the option value decays with time), and $\Gamma > 0$ (at a fixed t, the option value is a convex function of x).

21. Also, $c(t,x) - x\,\partial_x c = -Ke^{-r(T-t)}N(d_-) < 0$, which means the delta hedger always needs to borrow money.

22. **Long Gamma Strategy:** Fix t and take the tangent to $c(t,\cdot)$ at the current price x_1,

$$Y(x) = \partial_x c(t,x_1)(x-x_1) + c(t,x_1).$$

By convexity ($\Gamma > 0$), for x in the neighbourhood of x_1,

$$c(t,x) - Y(x) > 0.$$

Thus a long position in the option, delta-hedged by the stock, benefits from a move of the stock in either direction. However, $\Theta < 0$ means the option value also decays with time, and the curve drops very rapidly, so this arbitrage may not be realised; it is only infinitesimal.

23. **Put-Call Parity** (motivated by the no-arbitrage theorem). A *forward contract* on a stock delivers the stock at a future time T at a fixed delivery price K. The payoff for a long position is $\mathbb{S}(T) - K$.

    (a) The no-arbitrage delivery price is $K = e^{rT}\mathbb{S}(0)$, called the forward price $\mathrm{For}(T)$. The argument is a *static hedge*: a short position in the forward can be hedged by buying the stock at time 0 by borrowing $\mathbb{S}(0)$ from the money market, and at time T delivering the stock at $K = e^{rT}\mathbb{S}(0)$, which exactly repays the loan.

    (b) Let $f(x,t)$ denote the value of the forward contract. To sell the contract at time $t_0 < T$, the hedger gains $f(\mathbb{S}(t_0),t_0)$, purchases the stock at $\mathbb{S}(t_0)$ by borrowing $\mathbb{S}(t_0) - f(\mathbb{S}(t_0),t_0)$. At $t = T$ the position must be worth zero after delivering at K,

$$e^{r(T-t_0)}\big(\mathbb{S}(t_0) - f(\mathbb{S}(t_0),t_0)\big) - K = 0.$$

    (c) Hence

$$f(\mathbb{S}(t),t) = \mathbb{S}(t) - e^{-r(T-t)}K = \mathbb{S}(t) - e^{rt}\mathbb{S}(0).$$

24. **Payoff identity:** A long forward has payoff $x-K$, a long European call has $(x-K)^+$ and a short put has $-(K-x)^+$. Clearly

$$(x-K) = (x-K)^+ - (K-x)^+.$$

Equivalently, $x = K + (x-K)^+ - (K-x)^+$: one share of stock equals a zero-coupon bond, a long call and a short put. Rearranging,

$$x + (K-x)^+ = K + (x-K)^+,$$

i.e. one share plus a long put equals a zero-coupon bond plus a long call.

25. By no-arbitrage pricing, the same relation holds for $t < T$:

$$f(x,t) = c(x,t) - p(x,t) = x - e^{-r(T-t)}K.$$

Substituting the Black-Scholes-Merton formula for c and using $1 - N(d_\pm) = N(-d_\pm)$, we obtain the European put price

$$p(x,t) = Ke^{-r(T-t)}N\big(-d_-(T-t,x)\big) - x\,N\big(-d_+(T-t,x)\big).$$

26. **Ito Integrals.** In ordinary calculus, the chain rule gives $\int_{g(x_1)}^{g(x_2)} F(x)\,dg(x) = \int_{x_1}^{x_2} F(x)\frac{dg(x)}{dx}\,dx$. For $F(\mathbb{W}(T)) - F(\mathbb{W}(0))$ we saw that the chain rule fails (an extra $\frac{1}{2}F''\,dt$ term appears), so we need a genuinely new definition of $\int_0^T G(t)\,d\mathbb{W}(t)$.

27. We approximate it by the Riemann-type sum

$$\int_0^T G(t)\,d\mathbb{W}(t) \simeq \sum_{j=1}^{n} G(c_j)\big(\mathbb{W}(t_j) - \mathbb{W}(t_{j-1})\big), \qquad t_{j-1} \le c_j \le t_j.$$

For an ordinary integral $\int_0^T G(t)\,dt = \lim\sum G(c_j)(t_j - t_{j-1})$, the choice of c_j does not matter. For a stochastic $d\mathbb{W}$ it *does*, since $\mathbb{W}$ has non-zero quadratic variation. Common choices:

    (a) **Ito's choice:** $c_j = t_{j-1}$ (left end point).

    (b) **Stratonovich choice:** $c_j = \frac{t_{j-1}+t_j}{2}$ (mid point).

    (c) **Anti-Ito choice:** $c_j = t_j$ (right end point).

We use Ito's choice. It ensures the quadratic variation of $\mathbb{W}(t)$ is transferred to that of the integral, and also that the strategy G is not anticipating (it uses only information up to t_{j-1}, so no arbitrage from peeking into the future).

28. **Ito Integral of a function w.r.t. the Wiener process:**

$$\mathbb{I}(t) = \int_0^t \Delta(u)\, d\mathbb{W}(u),$$

where $\Delta(u)$ may be an ordinary function of time, or a random process, which is usually adapted to the filtration $\mathcal{F}(t)$ of the Wiener process. $\mathbb{I}(t)$ is itself a random process since it depends on t and is a sum of random variables.

29. **Construction:** We cannot use the chain rule, so we use the Riemann form with simple (step-function) processes. Define the simple process $\Delta_n(t) = \Delta(t_{j-1})$ for $t \in [t_{j-1}, t_j)$. Then $\Delta_n(t) \to \Delta(t)$ with

$$\lim_{n\to\infty} \mathbb{E}\left[\int_0^T |\Delta_n(t) - \Delta(t)|^2\, dt\right] = 0,$$

and

$$\mathbb{I} = \lim_{\substack{n\to\infty \\ \|\Pi\|\to 0}} \sum_{j=1}^{n} \Delta_n(t_{j-1})\big(\mathbb{W}(t_j) - \mathbb{W}(t_{j-1})\big).$$

30. **Properties of Ito Integrals.** For $\mathbb{I}(t) = \int_0^t \Delta(u)\, d\mathbb{W}(u)$:

    (a) Paths of $\mathbb{I}(t)$ are continuous.

    (b) $\mathbb{I}(t)$ is adapted to the filtration of $\mathbb{W}(t)$, since $\mathbb{W}(t_j) - \mathbb{W}(t_{j-1})$ is independent of the path up to t_{j-1}.

    (c) **Linearity:** if $\mathbb{J}(t) = \int_0^t \Gamma(u)\, d\mathbb{W}(u)$, then $\int_0^t (c_1\Delta(u) + c_2\Gamma(u))\, d\mathbb{W}(u) = c_1\mathbb{I}(t) + c_2\mathbb{J}(t)$.

    (d) **Martingale:** for $s < t$, $\mathbb{E}[\mathbb{I}(t) \mid \mathcal{F}(s)] = \mathbb{I}(s)$ (and hence $\mathbb{E}[\mathbb{I}(t)] = 0$). This is the most important property.

    (e) **Ito isometry:** $\mathbb{E}[\mathbb{I}(t)^2] = \mathrm{Var}(\mathbb{I}(t)) = \mathbb{E}\left[\int_0^t \Delta(u)^2\, du\right]$.

    (f) **Quadratic variation:** $[\mathbb{I}, \mathbb{I}](t) = \int_0^t \Delta(u)^2\, du$.

    Properties (b), (d), (e), (f) depend crucially on Ito's choice of c_j.

31. **Differential form:** $d\mathbb{I}(t) = \Delta(t)\, d\mathbb{W}(t)$, with $d\mathbb{I}\, d\mathbb{I} = \Delta(t)^2\, dt$.

32. **Ito Process:** Let $\Delta(t)$ and $\Theta(t)$ be stochastic processes adapted to the filtration $\mathcal{F}(t)$ of the Brownian motion. An Ito process is

$$\mathbb{X}(t) = \mathbb{X}(0) + \int_0^t \Delta(s)\, d\mathbb{W}(s) + \int_0^t \Theta(s)\, ds,$$

or in differential form

$$d\mathbb{X}(t) = \Delta(t)\, d\mathbb{W}(t) + \Theta(t)\, dt.$$

The first term is the diffusion, the second the drift. The delta hedging wealth equation $d\mathbb{X}(t) = \Delta(t)\, d\mathbb{S}(t) + r(\mathbb{X}(t) - \Delta(t)\mathbb{S}(t))\, dt$ is an example.

33. **Properties of Ito processes:**

    (a) An Ito process is *not necessarily* a martingale, due to the Θ term.

    (b) $[\mathbb{X}, \mathbb{X}](t) = \int_0^t \Delta(u)^2\, du$, i.e. $d\mathbb{X}\, d\mathbb{X} = \Delta(t)^2\, dt$ (the drift term does not contribute).

    (c) **Integral w.r.t. an Ito process:**

$$\int_0^t \Gamma(u)\, d\mathbb{X}(u) = \int_0^t \Gamma(u)\Delta(u)\, d\mathbb{W}(u) + \int_0^t \Gamma(u)\Theta(u)\, du.$$

34. **Ito-Doeblin Formula for a general Ito process.** Recall that for $f(\mathbb{W}(t))$,

$$f(\mathbb{W}(t)) - f(\mathbb{W}(0)) = \underbrace{\int_0^t f'(\mathbb{W}(u))\, d\mathbb{W}(u)}_{\text{diffusion}} + \underbrace{\frac{1}{2}\int_0^t f''(\mathbb{W}(u))\, du}_{\text{drift}},$$

which is an Ito process with $\Delta(t) = f'(\mathbb{W}(t))$ and $\Theta(t) = \frac{1}{2}f''(\mathbb{W}(t))$. For a general Ito process $\mathbb{X}(t)$,

$$\boxed{df(\mathbb{X}(t)) = f'(\mathbb{X}(t))\, d\mathbb{X}(t) + \frac{1}{2}f''(\mathbb{X}(t))\, d\mathbb{X}(t)\, d\mathbb{X}(t)}$$

Expanding using $d\mathbb{X} = \Delta\, d\mathbb{W} + \Theta\, dt$ and $d\mathbb{X}\, d\mathbb{X} = \Delta^2\, dt$,

$$df(\mathbb{X}(t)) = f'(\mathbb{X}(t))\Delta(t)\, d\mathbb{W}(t) + \left[f'(\mathbb{X}(t))\Theta(t) + \frac{1}{2}f''(\mathbb{X}(t))\Delta(t)^2\right] dt.$$

35. **Summary (things to remember):**

- Wiener process: $\mathbb{W}(t)$ is a martingale with $[\mathbb{W},\mathbb{W}](t) = t$, $d\mathbb{W}\, d\mathbb{W} = dt$.
- Ito integral: $\mathbb{I}(t) = \int_0^t \Delta(s)\, d\mathbb{W}(s)$ with Δ adapted; it is a martingale, $d\mathbb{I} = \Delta\, d\mathbb{W}$, $[\mathbb{I},\mathbb{I}](t) = \int_0^t \Delta(s)^2\, ds$.
- Ito process: $d\mathbb{X} = \Delta\, d\mathbb{W} + \Theta\, dt$; not a martingale, $d\mathbb{X}\, d\mathbb{X} = \Delta^2\, dt$.
- Ito-Doeblin: $df(\mathbb{X}) = f'(\mathbb{X})\, d\mathbb{X} + \frac{1}{2} f''(\mathbb{X})\, d\mathbb{X}\, d\mathbb{X}$.

4 Change of Measure

1. **Setup:** A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F}$ is a σ-algebra (a collection of subsets of the power set of Ω) and $\mathbb{P}$ the probability measure. A change of measure keeps $(\Omega, \mathcal{F})$ fixed and changes only $\mathbb{P}$. For instance, in the coin toss model with probabilities $p, q > 0$, $p + q = 1$, and

$$\mathbb{S}_{n+1} = \begin{cases} u\,\mathbb{S}_n, & \omega_{n+1} = H \\ d\,\mathbb{S}_n, & \omega_{n+1} = T \end{cases} \qquad (u > 1 > d),$$

the *empirical* (real world) probabilities p, q are inferred from observation.

2. **Why change measure?** Consider a hedging strategy for a short position in an option, based on $\mathbb{S}_n$, with portfolio $\mathbb{X}_{n+1} = \Delta_n \mathbb{S}_{n+1} + (1 + r)(\mathbb{X}_n - \Delta_n \mathbb{S}_n)$. We demand that $\mathbb{X}_n$ has mean return equal to the risk-free rate r, i.e. $\mathbb{E}[\mathbb{X}_{n+1} \mid \mathcal{F}_n] = (1 + r)\mathbb{X}_n$. Taking conditional expectations of the wealth equation, the $\mathbb{X}_n - \Delta_n \mathbb{S}_n$ terms cancel and we are left with

$$\mathbb{E}[\mathbb{S}_{n+1} \mid \mathcal{F}_n] = (1 + r)\mathbb{S}_n.$$

This imposes a constraint on p, q, r: $up + dq = 1 + r$ and $p + q = 1$, giving

$$\tilde{p} = \frac{1 + r - d}{u - d}, \qquad \tilde{q} = \frac{u - (1 + r)}{u - d},$$

called the *risk-neutral measure*. The empirical p, q generally do not satisfy this, so we need a precise way to move from $\mathbb{P}$ to $\tilde{\mathbb{P}}$.

3. **Radon-Nikodym Derivative (finite Ω).** Expectations under the two measures are

$$\mathbb{E}[\mathbb{X}] = \sum_{\omega \in \Omega} \mathbb{X}(\omega)\mathbb{P}\{\omega\}, \qquad \tilde{\mathbb{E}}[\mathbb{X}] = \sum_{\omega \in \Omega} \mathbb{X}(\omega)\tilde{\mathbb{P}}\{\omega\} = \sum_{\omega \in \Omega} \mathbb{X}(\omega)\frac{\tilde{\mathbb{P}}\{\omega\}}{\mathbb{P}\{\omega\}}\mathbb{P}\{\omega\}.$$

Define

$$\mathbb{Z}(\omega) = \frac{\tilde{\mathbb{P}}\{\omega\}}{\mathbb{P}\{\omega\}} \quad \left(\frac{\text{new}}{\text{old}}\right),$$

the *Radon-Nikodym derivative*. Then

$$\tilde{\mathbb{E}}[\mathbb{X}] = \mathbb{E}[\mathbb{X}\mathbb{Z}],$$

the expectation under the new measure is the expectation under the old measure, but weighted by the RN derivative.

4. **Properties of the RN derivative:**

   (a) $\mathbb{P}\{\mathbb{Z} > 0\} = 1$ (almost surely), since $\tilde{\mathbb{P}}\{\omega\}$ must be positive wherever $\mathbb{P}\{\omega\}$ is.

   (b) $\mathbb{E}[\mathbb{Z}] = \sum_\omega \frac{\tilde{\mathbb{P}}\{\omega\}}{\mathbb{P}\{\omega\}}\mathbb{P}\{\omega\} = \sum_\omega \tilde{\mathbb{P}}\{\omega\} = 1$.

   (c) $\tilde{\mathbb{E}}[\mathbb{X}] = \mathbb{E}[\mathbb{X}\mathbb{Z}]$.

   Any RN derivative must satisfy all three.

5. **Example:** $N = 2$ period binomial model with $\mathbb{S}_0 = 4$, $u = 2$, $d = 1/2$, $r = 1/4$. Then $\tilde{p} = \tilde{q} = 1/2$. Take the empirical $p = 2/3$, $q = 1/3$ and $\Omega = \{HH, HT, TH, TT\}$:

$$\mathbb{Z}(HH) = \frac{\tilde{p}^2}{p^2} = \frac{1/4}{4/9} = \frac{9}{16}, \qquad \mathbb{Z}(HT) = \mathbb{Z}(TH) = \frac{\tilde{p}\tilde{q}}{pq} = \frac{1/4}{2/9} = \frac{9}{8}, \qquad \mathbb{Z}(TT) = \frac{\tilde{q}^2}{q^2} = \frac{1/4}{1/9} = \frac{9}{4}.$$

   As a check of (b): $\frac{4}{9} \cdot \frac{9}{16} + 2 \cdot \frac{2}{9} \cdot \frac{9}{8} + \frac{1}{9} \cdot \frac{9}{4} = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$.

6. **Option pricing with the RN derivative.** For a European call with $\mathbb{V}_N = (\mathbb{S}_N - K)^+$, the fair value at time 0 is

$$\mathbb{V}_0 = \tilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1 + r)^N}\right] = \mathbb{E}\left[\frac{\mathbb{V}_N \mathbb{Z}}{(1 + r)^N}\right],$$

   and at time n,

$$\mathbb{V}_n = \tilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1 + r)^{N-n}} \,\middle|\, \mathcal{F}_n\right].$$

The factor $\frac{\mathbb{Z}}{(1+r)^N}$ appears in all formulae involving a hedging portfolio containing a money market component. We name it the *state price density* (a random variable),

$$\zeta(\omega) = \frac{\mathbb{Z}(\omega)}{(1+r)^N}, \qquad \mathbb{V}_0 = \mathbb{E}[\mathbb{V}_N \zeta].$$

It is the *balancing factor*: it rebalances the empirical probability of each path and accounts for the time value of money through the discounting factor.

7. **State prices:** Consider an instrument which pays 1 only when $\omega = \omega_0$ and 0 otherwise. Its price is

$$\mathbb{V}_0 = \mathbb{V}_N(\omega_0)\frac{\mathbb{Z}(\omega_0)}{(1+r)^N}\mathbb{P}\{\omega_0\} = \zeta(\omega_0)\mathbb{P}\{\omega_0\},$$

and $\zeta(\omega_0)\mathbb{P}\{\omega_0\}$ is called the *state price* of ω_0. Note $\mathbb{V}_0 = \mathbb{E}[\mathbb{V}_N\zeta]$ is under the empirical measure.

8. **Radon-Nikodym Derivative Process.** Define

$$\mathbb{Z}_n = \mathbb{E}[\mathbb{Z} \mid \mathcal{F}_n], \qquad n = 0, 1, \ldots, N,$$

where $\mathcal{F}_n$ is the σ-algebra generated by the first n coin tosses. $\mathbb{Z}_n$ is a random variable on Ω_N that depends only on $\omega_1, \ldots, \omega_n$, so it is an adapted process. It is a martingale w.r.t. $\{\mathcal{F}_n\}$: $\mathbb{E}[\mathbb{Z}_n \mid \mathcal{F}_m] = \mathbb{Z}_m$ for $m \le n$ (tower property). This is the *Radon-Nikodym derivative process*.

9. If $\mathbb{X}$ is $\mathcal{F}_m$-measurable, $m \le N$, then $\tilde{\mathbb{E}}[\mathbb{X}] = \mathbb{E}[\mathbb{X}\mathbb{Z}_m]$.

Proof.

$$\tilde{\mathbb{E}}[\mathbb{X}] = \mathbb{E}[\mathbb{X}\mathbb{Z}] = \mathbb{E}\big[\mathbb{E}[\mathbb{X}\mathbb{Z} \mid \mathcal{F}_m]\big] = \mathbb{E}\big[\mathbb{X}\,\mathbb{E}[\mathbb{Z} \mid \mathcal{F}_m]\big] = \mathbb{E}[\mathbb{X}\mathbb{Z}_m]$$

where we pulled $\mathbb{X}$ out since it is $\mathcal{F}_m$-measurable. □

10. **Closed form for the binomial model.** Splitting the sum over the first m tosses (which are fixed) and the remaining $m+1, \ldots, N$,

$$\mathbb{Z}_m = \sum_{\omega_{m+1}\ldots\omega_N} \mathbb{Z}(\omega_1 \ldots \omega_N)\, p^{n_H(\omega_{m+1}\ldots\omega_N)} q^{n_T(\omega_{m+1}\ldots\omega_N)} = \left(\frac{\tilde{p}}{p}\right)^{n_H(\omega_1\ldots\omega_m)} \left(\frac{\tilde{q}}{q}\right)^{n_T(\omega_1\ldots\omega_m)} \sum_{\omega_{m+1}\ldots\omega_N} \tilde{p}^{\,n_H}\tilde{q}^{\,n_T}.$$

The last sum is the total probability under $\tilde{\mathbb{P}}$, equal to 1. Hence

$$\mathbb{Z}_m = \left(\frac{\tilde{p}}{p}\right)^{n_H(\omega_1\ldots\omega_m)} \left(\frac{\tilde{q}}{q}\right)^{n_T(\omega_1\ldots\omega_m)}.$$

11. **Conditional expectation under the new measure (Bayes' rule).** If $\mathbb{X}$ is $\mathcal{F}_m$-measurable and $n \le m$, then

$$\tilde{\mathbb{E}}[\mathbb{X} \mid \mathcal{F}_n] = \frac{1}{\mathbb{Z}_n}\mathbb{E}[\mathbb{X}\mathbb{Z}_m \mid \mathcal{F}_n]. \qquad (\clubsuit)$$

(Proof: for any $\mathcal{F}_n$-measurable $\mathbb{Y}$, $\tilde{\mathbb{E}}[\mathbb{Y}\,\tilde{\mathbb{E}}[\mathbb{X} \mid \mathcal{F}_n]] = \tilde{\mathbb{E}}[\mathbb{Y}\mathbb{X}] = \mathbb{E}[\mathbb{Y}\mathbb{X}\mathbb{Z}_m] = \mathbb{E}\big[\mathbb{Y}\,\mathbb{E}[\mathbb{X}\mathbb{Z}_m \mid \mathcal{F}_n]\big] = \tilde{\mathbb{E}}\big[\mathbb{Y}\,\frac{1}{\mathbb{Z}_n}\mathbb{E}[\mathbb{X}\mathbb{Z}_m \mid \mathcal{F}_n]\big]$.)

12. **Pricing at intermediate times.** Using ($\clubsuit$),

$$\mathbb{V}_n = (1+r)^n\,\tilde{\mathbb{E}}\left[\frac{\mathbb{V}_N}{(1+r)^N}\,\middle|\,\mathcal{F}_n\right] = (1+r)^n\frac{1}{\mathbb{Z}_n}\mathbb{E}\left[\frac{\mathbb{V}_N\mathbb{Z}_N}{(1+r)^N}\,\middle|\,\mathcal{F}_n\right] = \frac{1}{\zeta_n}\mathbb{E}[\mathbb{V}_N\zeta_N \mid \mathcal{F}_n],$$

where $\zeta_n = \mathbb{Z}_n/(1+r)^n$. In summary,

$$\boxed{\mathbb{V}_0 = \mathbb{E}[\mathbb{V}_N\zeta_N], \qquad \mathbb{V}_n = \frac{1}{\zeta_n}\mathbb{E}[\mathbb{V}_N\zeta_N \mid \mathcal{F}_n]}$$

13. **Radon-Nikodym Derivative for Uncountable Sample Spaces.** For a finite Ω the expectation is a sum over ω (a *Riemann* sum over the domain). We can instead sum over the range of $\mathbb{X}$ (a *Lebesgue* sum over the range, which is a profound shift):

$$\mathbb{E}[\mathbb{X}] = \sum_{\omega\in\Omega}\mathbb{X}(\omega)\mathbb{P}\{\omega\} = \sum_{v\in\mathrm{Ran}(\mathbb{X})} v\,\mathbb{P}\{\mathbb{X} = v\}.$$

For a general sample space this becomes

$$\mathbb{E}[\mathbb{X}] = \int_\Omega \mathbb{X}(\omega)\, d\mathbb{P}(\omega) = \int_{\mathbb{R}} v\, d\nu_{\mathbb{X}}(v) = \int_{\mathbb{R}} v\, \phi_{\mathbb{X}}(v)\, dv,$$

where $d\nu_{\mathbb{X}}(v) = \mathbb{P}\{v \le \mathbb{X}(\omega) \le v + dv\}$ is the distribution of $\mathbb{X}$ and $\phi_{\mathbb{X}}$ its probability density. More generally, $\mathbb{E}[f(\mathbb{X})] = \int_{\mathbb{R}} f(v)\, d\nu_{\mathbb{X}}(v) = \int_{\mathbb{R}} f(v)\phi_{\mathbb{X}}(v)\, dv$.

14. For a continuous sample space we choose a random variable $\mathbb{Z}$ such that

    (a) $\mathbb{E}[\mathbb{Z}] = 1$,
    (b) $\mathbb{P}\{\mathbb{Z} > 0\} = 1$,

    and define the new probability measure on $(\Omega, \mathcal{F})$ by

    $$\boxed{\tilde{\mathbb{P}}(A) = \int_A \mathbb{Z}(\omega)\, d\mathbb{P}(\omega)}$$

    (1) implies $\tilde{\mathbb{P}}(\Omega) = 1$. Then $\tilde{\mathbb{E}}[\mathbb{X}] = \int_\Omega \mathbb{X}\, d\tilde{\mathbb{P}} = \int_\Omega \mathbb{X}\mathbb{Z}\, d\mathbb{P} = \mathbb{E}[\mathbb{X}\mathbb{Z}]$, and formally

    $$\mathbb{Z}(\omega) = \frac{d\tilde{\mathbb{P}}(\omega)}{d\mathbb{P}(\omega)}.$$

    The two ways of computing $\tilde{\mathbb{E}}[\mathbb{X}]$ are: $\int_{\mathbb{R}} v\, d\tilde{\nu}_{\mathbb{X}}(v)$ (distribution of $\mathbb{X}$ under $\tilde{\mathbb{P}}$), or $\int_{\mathbb{R}} v\,\zeta\, d\nu_{\mathbb{X}\mathbb{Z}}(v,\zeta)$ (joint distribution under $\mathbb{P}$).

15. **Example:** $\Omega = [0,1]$, $\mathcal{F}$ the Borel σ-algebra, $\mathbb{P}$ the Lebesgue measure ($\mathbb{P}[a,b] = b - a$), and $\mathbb{X}(\omega) = \omega$. Then $\mathbb{E}[\mathbb{X}] = \int_0^1 v\, dv = 1/2$. Take $\mathbb{Z}(\omega) = 2\omega$: $\mathbb{E}[\mathbb{Z}] = \int_0^1 2x\, dx = 1$ and $\mathbb{Z} > 0$ almost surely, so

    $$\tilde{\mathbb{P}}([a,b]) = \int_a^b 2\omega\, d\omega = b^2 - a^2.$$

    The strict inequality $\mathbb{Z} > 0$ is optional in general, but if you want to go back and forth between $\mathbb{P}$ and $\tilde{\mathbb{P}}$, the inequality must be strict.

16. **Equivalent measures:** $\mathbb{P}$ and $\tilde{\mathbb{P}} = \int \mathbb{Z}\, d\mathbb{P}$ are *equivalent* probability measures on $(\Omega, \mathcal{F})$, i.e. they agree on which events have probability 0.

17. **Radon-Nikodym Theorem:** If $\mathbb{P}$ and $\mathbb{P}'$ are two equivalent probability measures, then they are necessarily related by an RN derivative.

18. **Gaussian example (shifting the mean).** Let $\mathbb{X} \sim N(0, \sigma^2)$ under $\mathbb{P}$ with density $\phi_{\mathbb{X}}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/(2\sigma^2)}$, and

    $$\mathbb{Z}(\omega) = \exp\left(-\theta\mathbb{X}(\omega) - \tfrac{1}{2}\theta^2\sigma^2\right).$$

    (a) $\mathbb{E}[\mathbb{Z}] = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x + \theta\sigma^2)^2\right) dx = 1$ by completing the square and substituting $y = x + \theta\sigma^2$.
    (b) The expectation under the new measure is

    $$\tilde{\mathbb{E}}[\mathbb{X}] = \mathbb{E}[\mathbb{X}\mathbb{Z}] = \int_{-\infty}^{\infty} (y - \theta\sigma^2) \frac{e^{-y^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}\, dy = -\theta\sigma^2,$$

    i.e. $\tilde{\mathbb{E}}[\mathbb{X} + \theta\sigma^2] = 0$.

    Under $\tilde{\mathbb{P}}$, $\mathbb{X}$ has density $\tilde{\phi}_{\mathbb{X}}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x+\theta\sigma^2)^2}{2\sigma^2}\right)$, so $\mathbb{X} \sim N(-\theta\sigma^2, \sigma^2)$: the new measure shifts the mean while keeping the variance. Indeed $\mathbb{Z}(x) = \frac{d\tilde{\phi}_{\mathbb{X}}(x)}{d\phi_{\mathbb{X}}(x)}$. (Exercise: what should the RN derivative be to change the variance?)

19. **Shifting the Wiener process.** $\mathbb{W}(t) \sim N(0, t)$. For each t, let

    $$\mathbb{Z}(t) = \exp\left(-\theta\mathbb{W}(t) - \tfrac{1}{2}\theta^2 t\right).$$

    Here $\sigma^2 = t$ in the Gaussian example, so the random variable

    $$\tilde{\mathbb{W}}(t) = \mathbb{W}(t) + \theta t$$

    is a standard Wiener process with respect to the new measure $\tilde{\mathbb{P}}$.

20. **Generalising to time-dependent θ.** Writing $\mathbb{Z}(t) = \exp\Big(-\theta\int_0^t d\mathbb{W}(u) - \frac{1}{2}\theta^2\int_0^t du\Big)$ (think of t as $[\mathbb{W},\mathbb{W}](t)$), we generalise to

$$\mathbb{Z}(t) = \exp\bigg(-\int_0^t \Theta(u)\,d\mathbb{W}(u) - \frac{1}{2}\int_0^t \Theta(u)^2\,du\bigg), \qquad \tilde{\mathbb{W}}(t) = \mathbb{W}(t) + \int_0^t \Theta(u)\,du,$$

where $\Theta(u)$ is an adapted process w.r.t. the filtration of $\mathbb{W}(t)$. As before $\mathbb{Z}(t) = \mathbb{E}[\mathbb{Z}(T) \mid \mathcal{F}(t)]$ is the RN derivative process.

21. **Girsanov's Theorem.** Let $\mathbb{W}(t)$ be a Wiener process with filtration $\mathcal{F}(t)$, $0 \le t \le T$, and $\Theta(t)$ an $\mathcal{F}(t)$-adapted process. Let $\mathbb{Z}(t)$ be defined as above. Then

    (a) $\mathbb{Z}(T)$ is an RN derivative,

    (b) $\tilde{\mathbb{W}}(t) = \mathbb{W}(t) + \int_0^t \Theta(u)\,du$ is a Wiener process with respect to the measure $\tilde{\mathbb{P}}$ induced by $\mathbb{Z}(T)$,

    (c) $\mathbb{Z}(t) = \mathbb{E}[\mathbb{Z}(T) \mid \mathcal{F}(t)]$ is an RN derivative process.

    We now fix $\Theta(t)$ by demanding that the discounted stock price process is a martingale.

22. **Stock price process.** Under the old measure, $\mathbb{S}(t) = \mathbb{S}_0\exp(\sigma\mathbb{W}(t) - \frac{1}{2}\sigma^2 t)$ is a martingale ($d\mathbb{S} = \sigma\mathbb{S}\,d\mathbb{W}$). We modify it to the *generalised GBM*

$$d\mathbb{S}(t) = \alpha(t)\mathbb{S}(t)\,dt + \sigma(t)\mathbb{S}(t)\,d\mathbb{W}(t), \qquad (\star)$$

where $\alpha(t)$ (drift, the mean rate of return) and $\sigma(t)$ (diffusion) are adapted stochastic processes. Its solution (apply Ito-Doeblin to $\log\mathbb{S}$, $d\log\mathbb{S} = (\alpha - \frac{1}{2}\sigma^2)dt + \sigma\,d\mathbb{W}$) is

$$\mathbb{S}(t) = \mathbb{S}(0)\exp\bigg(\int_0^t \sigma(u)\,d\mathbb{W}(u) + \int_0^t \big(\alpha(u) - \tfrac{1}{2}\sigma(u)^2\big)\,du\bigg).$$

23. **Discount process.** Let the risk-free rate $\mathbb{R}(t)$ be an adapted process and

$$\mathbb{D}(t) = e^{-\int_0^t \mathbb{R}(u)\,du}, \qquad d\mathbb{D}(t) = -\mathbb{R}(t)\mathbb{D}(t)\,dt.$$

24. **Discounted stock price process.** By Ito's product formula, $d(\mathbb{D}\mathbb{S}) = (d\mathbb{D})\mathbb{S} + \mathbb{D}(d\mathbb{S}) + d\mathbb{D}\,d\mathbb{S}$. Substituting and retaining only terms of order dt and $d\mathbb{W}$ (the cross term vanishes),

$$d(\mathbb{D}(t)\mathbb{S}(t)) = (\alpha(t) - \mathbb{R}(t))\mathbb{D}(t)\mathbb{S}(t)\,dt + \sigma(t)\mathbb{D}(t)\mathbb{S}(t)\,d\mathbb{W}(t),$$

which is not necessarily a martingale. Use $d\mathbb{W}(t) = d\tilde{\mathbb{W}}(t) - \Theta(t)\,dt$:

$$d(\mathbb{D}\mathbb{S}) = (\alpha - \mathbb{R} - \sigma\Theta)\mathbb{D}\mathbb{S}\,dt + \sigma\mathbb{D}\mathbb{S}\,d\tilde{\mathbb{W}}.$$

25. **Choice of Θ:** To kill the drift, choose

$$\boxed{\Theta(t) = \frac{\alpha(t) - \mathbb{R}(t)}{\sigma(t)}}$$

called the *market price of risk*. Then

$$d(\mathbb{D}(t)\mathbb{S}(t)) = \sigma(t)\mathbb{D}(t)\mathbb{S}(t)\,d\tilde{\mathbb{W}}(t),$$

an Ito integral w.r.t. $\tilde{\mathbb{W}}$ and hence a martingale under $\tilde{\mathbb{P}}$. Its solution is

$$\mathbb{D}(t)\mathbb{S}(t) = \mathbb{S}(0)\exp\bigg(\int_0^t \sigma(u)\,d\tilde{\mathbb{W}}(u) - \frac{1}{2}\int_0^t \sigma(u)^2\,du\bigg), \qquad \mathbb{S}(t) = \mathbb{S}(0)\exp\bigg(\int_0^t \sigma\,d\tilde{\mathbb{W}} + \int_0^t \big(\mathbb{R} - \tfrac{1}{2}\sigma^2\big)\,du\bigg).$$

Under $\tilde{\mathbb{P}}$, the stock grows at the risk-free rate; the drift α has disappeared.

26. **Wealth process.** For $d\mathbb{X}(t) = \Delta(t)\,d\mathbb{S}(t) + \mathbb{R}(t)(\mathbb{X}(t) - \Delta(t)\mathbb{S}(t))\,dt$, we get

$$d(\mathbb{D}(t)\mathbb{X}(t)) = \Delta(t)\sigma(t)\mathbb{D}(t)\mathbb{S}(t)\,d\tilde{\mathbb{W}}(t) = \Delta(t)\,d(\mathbb{D}(t)\mathbb{S}(t)),$$

so $\mathbb{D}(t)\mathbb{X}(t)$ is a martingale under $\tilde{\mathbb{P}}$:

$$\mathbb{D}(t)\mathbb{X}(t) = \tilde{\mathbb{E}}[\mathbb{D}(T)\mathbb{X}(T) \mid \mathcal{F}(t)].$$

27. **Risk-neutral pricing formula.** We fix the fair price of an option to be $\mathbb{V}(t) = \mathbb{X}(t)$ for all $t \in [0,T]$ (the value of the replicating portfolio), so that

$$\boxed{\mathbb{D}(t)\mathbb{V}(t) = \tilde{\mathbb{E}}\big[\mathbb{D}(T)\mathbb{V}(T) \mid \mathcal{F}(t)\big]}$$

For a European call with strike K and expiration T, $\mathbb{V}(T) = (\mathbb{S}(T) - K)^+$.

28. **Black-Scholes-Merton formula via change of measure.** Specialise to constant α, σ, r, so $\mathbb{D}(t) = e^{-rt}$ and

$$\mathbb{S}(T) = \mathbb{S}(0)\exp\Big(\sigma\tilde{\mathbb{W}}(T) + \big(r - \tfrac{1}{2}\sigma^2\big)\,T\Big) = \mathbb{S}(t)\exp\Big(\sigma(\tilde{\mathbb{W}}(T) - \tilde{\mathbb{W}}(t)) + \big(r - \tfrac{1}{2}\sigma^2\big)\,(T-t)\Big).$$

Write $\tau = T - t$ (time to expiration) and $x = \mathbb{S}(t)$. Then

$$\mathbb{V}(t) = \tilde{\mathbb{E}}\left[e^{-r\tau}\left(x\exp\Big(\sigma(\tilde{\mathbb{W}}(T) - \tilde{\mathbb{W}}(t)) + \big(r - \tfrac{1}{2}\sigma^2\big)\,\tau\Big) - K\right)^+\right].$$

Define $\mathbb{Y} = -\frac{\tilde{\mathbb{W}}(T)-\tilde{\mathbb{W}}(t)}{\sqrt{\tau}} \sim N(0,1)$ under $\tilde{\mathbb{P}}$, with CDF $N(y) = \int_{-\infty}^{y} \frac{e^{-z^2/2}}{\sqrt{2\pi}}dz$. Then

$$\mathbb{V}(t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-r\tau}\left(x\,e^{-\sigma\sqrt{\tau}y + (r-\frac{1}{2}\sigma^2)\tau} - K\right)^+ e^{-y^2/2}\,dy.$$

29. The integrand is non-zero only when $x\,e^{-\sigma\sqrt{\tau}y+(r-\frac{1}{2}\sigma^2)\tau} > K$, i.e.

$$y < \frac{1}{\sigma\sqrt{\tau}}\left[\ln\frac{x}{K} + \big(r - \tfrac{1}{2}\sigma^2\big)\,\tau\right] = d_-(\tau, x).$$

Splitting the integral into the two terms,

$$\mathbb{V}(t) = \frac{x}{\sqrt{2\pi}}\int_{-\infty}^{d_-} e^{-\frac{1}{2}(y+\sigma\sqrt{\tau})^2}\,dy - \frac{Ke^{-r\tau}}{\sqrt{2\pi}}\int_{-\infty}^{d_-} e^{-\frac{1}{2}y^2}\,dy.$$

Substituting $z = y + \sigma\sqrt{\tau}$ in the first integral shifts its upper limit to $d_- + \sigma\sqrt{\tau} = d_+$, so

$$\boxed{\mathbb{V}(t) = x\,N\big(d_+(\tau,x)\big) - Ke^{-r\tau}N\big(d_-(\tau,x)\big)}$$

with

$$d_\pm(\tau,x) = \frac{1}{\sigma\sqrt{\tau}}\left[\ln\frac{x}{K} + \big(r \pm \tfrac{1}{2}\sigma^2\big)\,\tau\right] = \frac{1}{\sigma\sqrt{\tau}}\ln\Big(\frac{x}{Ke^{-r\tau}}\Big) \pm \frac{\sigma\sqrt{\tau}}{2}.$$

This is the same Black-Scholes-Merton formula we found by solving the PDE, now obtained as an expectation under the risk-neutral measure. The drift α never appears: it was traded for the market price of risk Θ in the change of measure.

5 Multi-dimensional Models and the Fundamental Theorems

1. **Multi-dimensional Wiener Process:** Many independent Wiener processes can drive the randomness in stock prices. A d-dimensional (vector valued) Wiener process is

$$\vec{\mathbb{W}}(t) = \big(\mathbb{W}_1(t), \mathbb{W}_2(t), \ldots, \mathbb{W}_d(t)\big),$$

where

   (a) each $\mathbb{W}_i$ is a Wiener process,

   (b) $\mathbb{W}_i$ and $\mathbb{W}_j$, $i \neq j$, are independent random variables, i.e. $\mathbb{P}\{\mathbb{W}_i(t) \in B, \mathbb{W}_j(t) \in B'\} = \mathbb{P}\{\mathbb{W}_i(t) \in B\}\mathbb{P}\{\mathbb{W}_j(t) \in B'\}$, equivalently $\sigma(\mathbb{W}_i)$ and $\sigma(\mathbb{W}_j)$ are independent.

2. **Filtration of $\vec{\mathbb{W}}$:** $\{\mathcal{F}(t)\}_{0\leq t\leq T}$ with $\mathcal{F}(s) \subseteq \mathcal{F}(t)$ for $s \leq t$, generated by the full collection of the $\mathbb{W}_i$, such that

   (a) $\vec{\mathbb{W}}(t)$ is $\mathcal{F}(t)$-measurable,

   (b) $\vec{\mathbb{W}}(t) - \vec{\mathbb{W}}(s)$ is independent of the information in $\mathcal{F}(s)$.

3. **Quadratic variations:** For each i, $d\mathbb{W}_i\, d\mathbb{W}_i = dt$, i.e. $[\mathbb{W}_i, \mathbb{W}_i](t) = t$. For $i \neq j$, $d\mathbb{W}_i\, d\mathbb{W}_j = 0$, i.e. $[\mathbb{W}_i, \mathbb{W}_j](t) = 0$ (due to independence).

4. **Ito process (single, driven by d Wiener processes):**

$$d\mathbb{X}(t) = \Theta(t)\, dt + \vec{\sigma}(t) \cdot d\vec{\mathbb{W}}(t),$$

with Θ and $\vec{\sigma}$ adapted.

5. **Collection of Ito processes:** For $\alpha = 1, 2, \ldots, n$,

$$d\mathbb{X}_\alpha(t) = \Theta_\alpha(t)\, dt + \sum_{i=1}^{d} \sigma_{\alpha i}(t)\, d\mathbb{W}_i(t).$$

For instance, with $n = 3$ processes and $d = 2$ Wiener processes,

$$\begin{pmatrix} d\mathbb{X}_1 \\ d\mathbb{X}_2 \\ d\mathbb{X}_3 \end{pmatrix} = \begin{pmatrix} \Theta_1 \\ \Theta_2 \\ \Theta_3 \end{pmatrix} dt + \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \\ \sigma_{31} & \sigma_{32} \end{pmatrix} \begin{pmatrix} d\mathbb{W}_1 \\ d\mathbb{W}_2 \end{pmatrix}.$$

In general $d\mathbb{X}$ and Θ are $n \times 1$ vectors, $d\vec{\mathbb{W}}$ is $d \times 1$, and σ is an $n \times d$ matrix. $\sigma_{\alpha i}$ is the "volatility" of $\mathbb{X}_\alpha$ corresponding to the Wiener process $\mathbb{W}_i$.

6. **Quadratic variation of a vector valued Ito process.** Retaining only the $d\mathbb{W}\, d\mathbb{W}$ terms,

$$d\mathbb{X}_\alpha\, d\mathbb{X}_\alpha = \sum_{i=1}^{d}\sum_{j=1}^{d} \sigma_{\alpha i}\sigma_{\alpha j}\, d\mathbb{W}_i\, d\mathbb{W}_j = \sum_{i=1}^{d} \sigma_{\alpha i}^2\, dt$$

since the terms with $i \neq j$ vanish. Similarly, the cross variation is

$$\boxed{d\mathbb{X}_\alpha\, d\mathbb{X}_\alpha = \left(\sum_{i=1}^{d} \sigma_{\alpha i}^2\right) dt, \qquad d\mathbb{X}_\alpha\, d\mathbb{X}_\beta = \left(\sum_{i=1}^{d} \sigma_{\alpha i}\sigma_{\beta i}\right) dt}$$

7. **Ito-Doeblin formula (multi-dimensional).** In one variable,

$$df(t,x) = \partial_t f\, dt + \partial_x f\, dx + \frac{1}{2}\partial_x^2 f\, dx\, dx.$$

For two variables,

$$df(t,x,y) = \partial_t f\, dt + \partial_x f\, dx + \partial_y f\, dy + \frac{1}{2}\partial_x^2 f\, dx\, dx + \frac{1}{2}\partial_y^2 f\, dy\, dy + \partial_x\partial_y f\, dx\, dy.$$

Note that there is no $\frac{1}{2}$ on the mixed term: the two cross terms $\frac{1}{2}\partial_x\partial_y f\, dx\, dy + \frac{1}{2}\partial_y\partial_x f\, dy\, dx$ combine because $\partial_x\partial_y f = \partial_y\partial_x f$ (true for "nice" functions). Likewise for three variables x, y, z, there are three squared terms with $\frac{1}{2}$ and three mixed terms $\partial_x\partial_y f\, dx\, dy + \partial_y\partial_z f\, dy\, dz + \partial_z\partial_x f\, dz\, dx$ without it.

8. **Girsanov's Theorem (d-dimensional).** Let $\vec{\mathbb{W}}(t)$ be a d-dimensional Wiener process with filtration $\{\mathcal{F}(t)\}_{0\le t\le T}$ and $\vec{\Theta}(t)$ a d-dimensional adapted process. Define

$$\mathbb{Z}(t) = \exp\left(-\int_0^t \vec{\Theta}(u)\cdot d\vec{\mathbb{W}}(u) - \frac{1}{2}\int_0^t \vec{\Theta}(u)\cdot\vec{\Theta}(u)\,du \right).$$

Then

(a) $\mathbb{Z}(T)$ is an RN derivative ($\mathbb{P}\{\mathbb{Z}(T)>0\}=1$, $\mathbb{E}[\mathbb{Z}(T)]=1$),

(b) $\mathbb{Z}(t)$ is an RN derivative process, with $\tilde{\mathbb{P}}(A) = \int_A \mathbb{Z}(T)\,d\mathbb{P}$,

(c) $\tilde{\vec{\mathbb{W}}}(t) = \vec{\mathbb{W}}(t) + \int_0^t \vec{\Theta}(u)\,du$ is a d-dimensional Wiener process under $\tilde{\mathbb{P}}$.

9. **Multi-dimensional Market Model.** A money market with risk-free rate $\mathbb{R}(t)$ and m stocks $\mathbb{S}_I$, $I=1,\dots,m$, driven by d Wiener processes:

$$d\mathbb{S}_I(t) = \alpha_I(t)\mathbb{S}_I(t)\,dt + \mathbb{S}_I(t)\sum_{j=1}^{d}\sigma_{Ij}(t)\,d\mathbb{W}_j(t),$$

where σ_{Ij} are the volatility coefficients. The discount process is $d\mathbb{D}(t) = -\mathbb{D}(t)\mathbb{R}(t)\,dt$. We want to find a fair price for a derivative security written on these m stocks for each $t\in[0,T]$, knowing the payoff at time T.

10. **Hedging portfolio and wealth process:** With $\Delta_I(t)$ shares of $\mathbb{S}_I$ and the rest in the money market,

$$d\mathbb{X}(t) = \sum_{I=1}^{m}\Delta_I(t)\,d\mathbb{S}_I(t) + \mathbb{R}(t)\left(\mathbb{X}(t) - \sum_{I=1}^{m}\Delta_I(t)\mathbb{S}_I(t)\right)dt.$$

11. **Is there a risk-neutral measure?** We need a measure $\tilde{\mathbb{P}}$ such that every discounted stock price $\mathbb{D}(t)\mathbb{S}_I(t)$ is a martingale, i.e. $\tilde{\mathbb{E}}[\mathbb{D}(t)\mathbb{S}_I(t)\mid\mathcal{F}(s)] = \mathbb{D}(s)\mathbb{S}_I(s)$ for $s<t$ (equivalently $\mathbb{E}[\mathbb{Z}\mathbb{D}(t)\mathbb{S}_I(t)\mid\mathcal{F}(s)] = \mathbb{Z}(s)\mathbb{D}(s)\mathbb{S}_I(s)$). Requirement: the discounted stock has no dt term under $\tilde{\mathbb{P}}$,

$$d(\mathbb{D}\mathbb{S}_I) = \mathbb{D}\mathbb{S}_I\sum_{j=1}^{d}\sigma_{Ij}\,d\tilde{\mathbb{W}}_j.$$

But what we know (by Ito's product formula) is

$$d(\mathbb{D}\mathbb{S}_I) = (\alpha_I-\mathbb{R})\mathbb{D}\mathbb{S}_I\,dt + \mathbb{D}\mathbb{S}_I\sum_{j=1}^{d}\sigma_{Ij}\,d\mathbb{W}_j.$$

Assume $d\tilde{\mathbb{W}}_j = d\mathbb{W}_j + \Theta_j\,dt$. Substituting, and comparing the dt terms with the requirement, we get the **market price of risk equations**:

$$\boxed{\alpha_I(t)-\mathbb{R}(t) = \sum_{j=1}^{d}\sigma_{Ij}(t)\Theta_j(t), \qquad I=1,\dots,m} \tag{$*$}$$

or in matrix form $(\alpha - \mathbb{R}\mathbf{1}) = \sigma\vec{\Theta}$, with σ the $m\times d$ matrix.

12. **Solutions of ($*$):** These are m linear equations in the d unknowns Θ_j. There are three possibilities:

(a) *No solution:* no $\tilde{\mathbb{P}}$ under which the $\mathbb{D}\mathbb{S}_I$ are martingales; this leads to arbitrage.

(b) *Unique solution.*

(c) *Infinitely many solutions.*

If there is at least one solution, a risk-neutral measure $\tilde{\mathbb{P}}$ exists (via Girsanov with that $\vec{\Theta}$).

13. **Arbitrage:** A wealth process $\mathbb{X}(t)$ such that

(a) $\mathbb{X}(0)=0$ (start with zero wealth),

(b) for some $T>0$, $\mathbb{P}\{\mathbb{X}(T)\ge 0\}=1$ (almost surely there is no loss of wealth),

(c) $\mathbb{P}\{\mathbb{X}(T)>0\}>0$ (some positive probability of gaining wealth).

14. **First Fundamental Theorem of Asset Pricing:** If the market price of risk equations $(*)$ have at least one solution (equivalently, if there is a risk-neutral measure $\tilde{\mathbb{P}}$), then there is no arbitrage.

    Proof. Suppose an arbitrage portfolio $\mathbb{X}(t)$ made of shares in the $\mathbb{S}_I$ and a money market account exists. Since $\tilde{\mathbb{P}}$ exists, $\mathbb{D}(t)\mathbb{X}(t)$ is a martingale under it, so
    $$\tilde{\mathbb{E}}[\mathbb{D}(T)\mathbb{X}(T)] = \mathbb{D}(0)\mathbb{X}(0) = 0. \tag{A}$$
    From $\mathbb{P}\{\mathbb{X}(T) \geq 0\} = 1$ and $\mathbb{D}(T) > 0$ we get $\mathbb{P}\{\mathbb{D}(T)\mathbb{X}(T) < 0\} = 0$, and by equivalence of $\mathbb{P}$ and $\tilde{\mathbb{P}}$ (they agree on probability-zero events), $\tilde{\mathbb{P}}\{\mathbb{D}(T)\mathbb{X}(T) < 0\} = 0$, hence
    $$\tilde{\mathbb{E}}[\mathbb{D}(T)\mathbb{X}(T)] \geq 0. \tag{B}$$
    If (A) and (B) hold simultaneously, then $\tilde{\mathbb{P}}\{\mathbb{D}(T)\mathbb{X}(T) > 0\} = 0$, and again by equivalence $\mathbb{P}\{\mathbb{X}(T) > 0\} = 0$. This contradicts requirement (c) for arbitrage. Hence no arbitrage exists. □

15. **Complete markets.** A (model of a) financial market is *complete* if every derivative security can be hedged: a short position on the security can be hedged, i.e. every derivative security can be replicated by a portfolio made up of the m stocks and the money market account.

16. **Second Fundamental Theorem of Asset Pricing:** A market model is complete if and only if there is a *unique* risk-neutral measure (i.e. a unique solution to the market price of risk equations).

17. **Martingale Representation Theorem:** (see Shreve II for the actual statement). If we have a martingale w.r.t. the risk-neutral measure $\tilde{\mathbb{P}}$, then it can always be represented as an Ito integral. In particular, for the discounted wealth,
    $$d(\mathbb{D}(t)\mathbb{X}(t)) = \sum_{j=1}^{d} \tilde{\Pi}_j(t)\, d\tilde{\mathbb{W}}_j(t),$$
    where the $\tilde{\Pi}_j$ are adapted processes.

18. **Why uniqueness is equivalent to completeness.** For a replicating portfolio, $d(\mathbb{D}\mathbb{X}) = \sum_I \Delta_I\, d(\mathbb{D}\mathbb{S}_I)$, so
    $$d(\mathbb{D}\mathbb{X}) = \sum_{I=1}^{m} \Delta_I \mathbb{D}\mathbb{S}_I \sum_{j=1}^{d} \sigma_{Ij}\, d\tilde{\mathbb{W}}_j = \mathbb{D} \sum_{j=1}^{d} \left(\sum_{I=1}^{m} \Delta_I \mathbb{S}_I \sigma_{Ij} \right) d\tilde{\mathbb{W}}_j.$$
    Comparing with the martingale representation,
    $$\boxed{\tilde{\Pi}_j(t) = \mathbb{D}(t) \sum_{I=1}^{m} \Delta_I(t)\mathbb{S}_I(t)\sigma_{Ij}(t), \qquad j = 1, \ldots, d}$$
    These are d equations in the m unknowns $\Delta_I(t)$. The market is complete iff these have a solution for every choice of $\tilde{\Pi}_j$ (i.e. every choice of portfolio/derivative), and this happens if and only if the market price of risk equations have a unique solution.

19. **Remark (linear algebra view).** Both conditions are statements about the rank of the $m \times d$ volatility matrix σ. $(*)$ has a unique solution $\vec{\Theta}$ (given that one exists) iff σ has d independent columns, and the hedging equations are solvable for every $\tilde{\Pi}$ iff σ^T is onto $\mathbb{R}^d$, which is again $\operatorname{rank}(\sigma) = d$. Intuitively: no arbitrage needs the stocks' risk premia to be consistent (solvability of $(*)$), and completeness needs at least as many independent traded stocks as there are independent sources of randomness ($m \geq d$ with full rank).